Tool-Calling LLM Agents

Just like they have back home in RL

- Traditional agents, given a goal:
  - **Plan** a sequence of actions
  - **Perceive** the state of the world
  - **Interact** with the external world using these actions

- With LLM agents:
  - Goal set by the user
  - Planning with Chain-of-Thought reasoning
  - Use tools to perceive and interact with the world

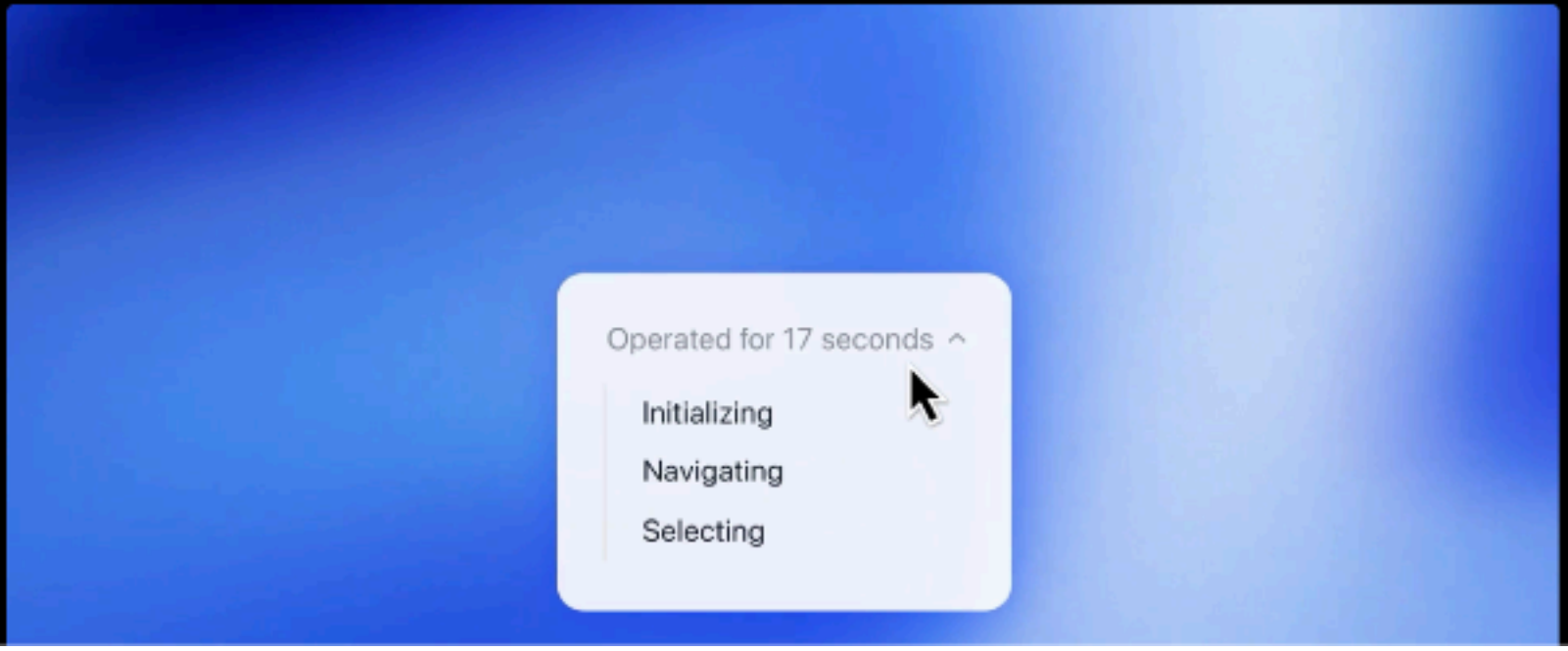
6

January 23, 2025 Release

Computer-Using Agent

Powering Operator with Computer-Using Agent, a universal interface for AI to interact with the digital world.

Go to Operator ↗



```
Claude Code v2.0.47

Welcome back Andrew!

Sonnet 4.5 · Claude Pro
~/Desktop/Code/py-playground

Tips for getting started
Run /init to create a CLAUDE.md file with instructions for …

Recent activity
No recent activity

> Try "create a util logging.py that..."

? for shortcuts                                Thinking on (tab to toggle)
```

Asta > Find papers

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andrz

Find papers about AI agents

Researched for 40 seconds View steps

Asta

I found 7 papers that look like perfect matches, 24 relevant ones and 31 others.

This is what I searched for:

- Papers discussing: *AI agents*.
- Judged by the following relevance criteria:
  - *AI Agents - Definition and Characteristics*: The paper defines or describes AI agents, including their key characteristics, properties, and components (e.g., perception, reasoning, action).
  - *AI Agents - Architectures and Frameworks*: The paper discusses architectures, frameworks, or platforms used to build or implement AI agents. This includes specific agent architectures (e.g., BDI, hierarchical), software frameworks, or hardware platforms.
  - *AI Agents - Applications and Use Cases*: The paper presents specific applications or use cases of AI agents in various domains (e.g., robotics, game playing, healthcare, finance). It should describe

62 Papers

Export All Citations

A Comprehensive Review of AI Agents: Transforming Possibilities in Technology and Beyond

Xiaodong Gu Andrews Damoah Joshua Sherwood +11 Authors Cristabelle Madona De Souza

• arXiv.org • 2025

Perfectly Relevant: This paper reviews AI agents, covering their definition, characteristics (perception, reasoning, action), architectures, and applications in real-world scenarios. It synthesizes insights from cognitive science-inspired models, hierarchical reinforcement learning frameworks, and large language model-based reasoning.

AI Agents - Definition and Characteristics

AI Agents - Architectures and Frameworks

AI Agents - Applications and Use Cases

Show Evidence

Cited by 2 Copy BibTeX

AI Agents: Evolution, Architecture, and Real-World Applications

Naveen Krishnan • arXiv.org • 2025

Perfectly Relevant: The paper defines AI agents by describing their evolution and components. It also reviews agent paradigms and analyzes applications across enterprise, personal assistance, and specialized domains.

AI Agents - Definition and Characteristics

Sort by

Relevance

(7) Perfectly Relevant

(24) Relevant

(31) Somewhat Relevant

Criteria

(49) AI Agents - Definition and Che...

(34) AI Agents - Applications and ...

(17) AI Agents - Architectures and ...

Year

Venue

Author

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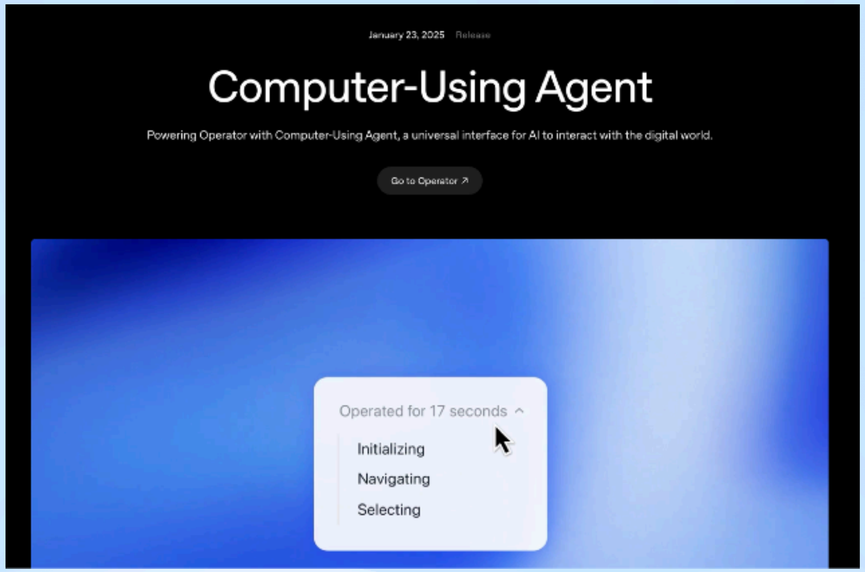

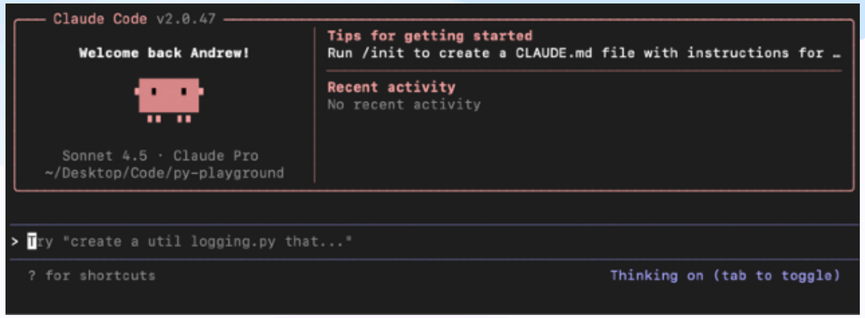

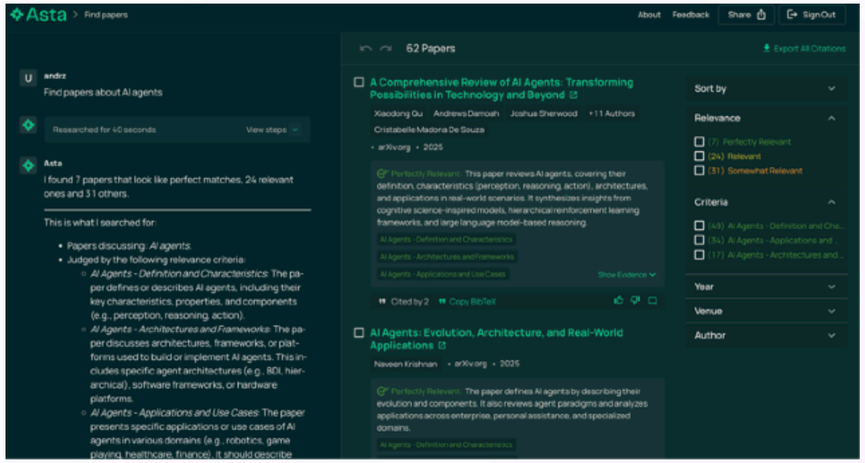

What properties are desirable in an agent framework in order to effectively build LLM agents that are effective and robust in both research and production?